

PF Fall2022, Assignment 2

Q1. In a right triangle, the square of the length of one side is equal to the sum of the squares of the length of the other two sides. Write a pseudo-code that takes in lengths of the three sides of a triangle, and then outputs whether the triangle is right or not.

Q2. Write a pseudo-code that takes in a number as input and prints whether it is negative, positive or zero.

Q3. Write a pseudo-code that takes in as input the number of units used by the consumer, and prints the total cost of electricity according to the given condition:

For first 110 units, the rate is Rs. 0.50/unit

For next 90 units, the rate is Rs. 0.80/unit

For unit above 200, the rate gets raised to Rs. 1.50/unit

Sample examples (these are for your understanding only, the pseudo-code must work for any value entered by the user):

Input: 100

Output: 50

How: $100 \times 0.5 = 50$.

Input: 111

Output: 55.8

How: $110 \times 0.5 + 1 \times 0.8 = 55 + 0.8 = 55.8$

Input: 300

Output: 277

How: $110 \times 0.5 + 90 \times 0.8 + 100 \times 1.5 = 55 + 72 + 150 = 277$